EE2-08A MATHEMATICS

1. (a) Show that the function $v(x,y) = e^{-y}\sin x - e^x \cos y$ satisfies Laplace's equation and integrate the Cauchy-Riemann equations to find its harmonic conjugate $u(x,y)$.

SOLUTION

First differentiate:

$v_x = e^{-y}\cos x - e^x \cos y \Rightarrow v_{xx} = -e^{-y}\sin x - e^x \cos y$ and

$v_y = -e^{-y}\sin x + e^x \sin y \Rightarrow v_{yy} = e^{-y}\sin x + e^x \cos y$ and hence $v_{xx} + v_{yy} = 0$ and v is a harmonic function, satisfying Laplace's equation. Therefore v has a harmonic conjugate u and together they satisfy the Cauchy-Riemann equations: $u_x = v_y$ and $u_y = -v_x$. Hence

$$u = \int v_y \, dx = \int -e^{-y}\sin x + e^x \sin y \, dx = e^{-y}\cos x + e^x \sin y + A(y)\,,$$

and

$$u = \int -v_x \, dy = -\int e^{-y}\cos x - e^x \cos y \, dy = e^{-y}\cos x + e^x \sin y + B(x)\,,$$

where A and B are arbitrary functions. Comparing the two, we see that $A = B =$constant, and

$$u(x,y) = e^{-y}\cos x + e^x \sin y + C$$

is the harmonic conjugate. [Seen similar.]

(b) Hence obtain $f(z) = u(x,y) + iv(x,y)$ where f is an analytic function of $z = x + iy$, simplifying as much as possible.

SOLUTION

Joining the two, we have

$$f(z) = u + iv = e^{-y}\cos x + e^x \sin y + i(e^{-y}\sin x - e^x \cos y) + C$$

$$= e^{-y}(\cos x + i\sin x) + e^x(\sin y - i\cos y) + C = e^{-y}e^{ix} - ie^x(\cos y + i\sin y) + C$$

$$= e^{-y+ix} - ie^x e^{iy} + C = e^{iz} - ie^z + C$$ [Seen similar.]

2. (a) The complex function

$$F(z) = \frac{e^{imz}}{(z^2+1)(z^2+4)}$$

has four simple poles. Find the residues at the poles lying in the upper half of the complex plane.

SOLUTION The poles are at $z^2 = -1, -4 \Rightarrow z = \pm i, \pm 2i$. Using the formula, we calculate the residue at $z = i$:

$$\text{Res}(z = i) = \lim_{z\to i}(z-i)F(z) = \lim_{z\to i}\frac{e^{imz}}{(z+i)(z^2+4)} = -i\frac{e^{-m}}{6},$$

and at $z = 2i$:

$$\text{Res}(z = 2i) = \lim_{z\to 2i}(z-2i)F(z) = \lim_{z\to 2i}\frac{e^{imz}}{(z^2+1)(z+2i)} = i\frac{e^{-2m}}{12}.$$

[Seen similar.]

(b) Consider the contour integral $I = \oint_C \frac{e^{imz}}{(z^2+1)(z^2+4)}\,dz$,

where the closed contour C consists of a semi-circle in the complex upper half-plane, taken in the anti-clockwise sense, and $m > 0$.

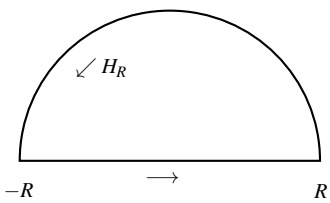
Using the result from (a), obtain

$$\int_{-\infty}^{\infty} \frac{\cos mx}{(x^2+1)(x^2+4)}\,dx,$$

explaining carefully your evaluation of I, and your use of Cauchy's Residue Theorem and Jordan's lemma.

SOLUTION

We use the contour C as the union of the semi-circular arc H_R in the upper half-plane and the real interval $[-R, R]$:



The contour integral can be written in two parts:

$$I = \oint_C \frac{e^{imz}}{(z^2+1)(z^2+4)}\,dz = \int_{H_R} \frac{e^{imz}}{(z^2+1)(z^2+4)}\,dz + \int_{-R}^{R} \frac{e^{imx}}{(x^2+1)(x^2+4)}\,dx \qquad (1)$$

Using Jordan's lemma

$$\lim_{R\to\infty} \int_{H_R} \frac{e^{imz}}{(z^2+4)^2}\,dz = 0\,,$$

because:
(i) $m > 0$,
(ii) $\left|\dfrac{1}{(z^2+1)(z^2+4)}\right| \to 0$ as $R \to \infty$,
(iii) all singularities are poles.

Using Cauchy's residue theorem,

$$I = \oint_C \frac{e^{imz}}{(z^2+1)(z^2+4)}\,dz = 2\pi i \times \text{ sum of residues in the upper half-plane}$$

$$= 2\pi i \left(-i\frac{e^{-m}}{6} + i\frac{e^{-2m}}{12}\right) = \pi\left(\frac{e^{-m}}{3} - \frac{e^{-2m}}{6}\right).$$

This holds as $R \to \infty$, as there are no further poles; taking the limit as $R \to \infty$, the other integral on the rhs of (1) yields

$$\int_{-\infty}^{\infty} \frac{e^{imx}}{(x^2+1)(x^2+4)}\,dx = \int_{-\infty}^{\infty} \frac{\cos mx + i\sin mx}{(x^2+1)(x^2+4)}\,dx = \int_{-\infty}^{\infty} \frac{\cos mx}{(x^2+1)(x^2+4)}\,dx = \pi\left(\frac{e^{-m}}{3} - \frac{e^{-2m}}{6}\right),$$

where the sine part vanishes as it's the symmetric integral of an odd function. [Seen similar.]

3. (a) If $g(a) \neq 0$, the complex function

$$f(z) = \frac{g(z)}{h(z)},$$

has a simple pole at $z = a$ if $h(z)$ has a simple zero at $z = a$. In that case, use a Taylor expansion of $h(z)$ to show that the residue of $f(z)$ at $z = a$ is

$$\text{Res}\,[f(z)]_{z=a} = \frac{g(a)}{h'(a)}.$$

SOLUTION Expanding $h(z) = h(a) + h'(a)(z-a) + \frac{1}{2}h''(a)(z-a)^2 + \cdots$ and noting that if a is a simple zero, then $h(a) = 0$ but $h'(a) \neq 0$, we write the residue as

$$\lim_{z\to a}(z-a)f(z) = \lim_{z\to a}\frac{(z-a)g(z)}{h'(a)(z-a)+\frac{1}{2}h''(a)(z-a)^2+\cdots} = \lim_{z\to a}\frac{g(z)}{h'(a)+\frac{1}{2}h''(a)(z-a)+\cdots} = \frac{g(a)}{h'(a)}$$

as required, given that all terms involving powers of $z-a$ vanish, leaving non-zero denominator and numerator. [Bookwork.]

(b) Show also, that if $g(z) = 1$, a constant, and $f(z)$ has a double pole at $z = a$, then the residue is

$$\text{Res}\,[f(z)]_{z=a} = -\frac{2h'''(a)}{3[h''(a)]^2}.$$

SOLUTION For a double pole we have $h(a) = h'(a) = 0$ with $h''(a) \neq 0$, and again we take the residue formula

$$\lim_{z\to a}\frac{d}{dz}\left[(z-a)^2 f(z)\right] = \lim_{z\to a}\frac{d}{dz}\left[\frac{(z-a)^2}{\frac{1}{2}h''(a)(z-a)^2+\frac{1}{6}h'''(a)(z-a)^3+\frac{1}{24}h''''(a)(z-a)^4+\cdots}\right]$$

$$= \lim_{z\to a}\frac{d}{dz}\left[\frac{1}{\frac{1}{2}h''(a)+\frac{1}{6}h'''(a)(z-a)+\frac{1}{24}h''''(a)(z-a)^2+\cdots}\right]$$

and using the chain rule:

$$= \lim_{z\to a}\left[-\left(\frac{1}{2}h''(a)+\frac{1}{6}h'''(a)(z-a)+\frac{1}{24}h''''(a)(z-a)^2+\cdots\right)^{-2}\left(\frac{1}{6}h'''(a)+\frac{1}{24}h''''(a)(z-a)+\cdots\right)\right]$$

Finally, all terms involing $z-a$ vanish, and as $h''(a) \neq 0$, we have a non-zero denominator:

$$= -\frac{\frac{1}{6}h'''(a)}{\left[\frac{1}{2}h''(a)\right]^2} = -\frac{2h'''(a)}{3[h''(a)]^2}, \text{ as required.}$$ [New.]

4. (a) A function $f(t)$ has Laplace transform $\mathscr{L}[f(t)]$. Prove the second shift-theorem, that for a positive constant a:

$$\mathscr{L}\left[H(t-a)f(t-a)\right] = e^{-as}\mathscr{L}[f(t)]\,,$$

where $H(t)$ is the unit-step, or Heaviside function.

SOLUTION

Let $u = t-a$. Then $t = 0 \Rightarrow u = -a$ and $dt = du$, so that

$$\mathscr{L}\left[H(t-a)f(t-a)\right] = \int_0^\infty e^{-st}H(t-a)f(t-a)\,dt = \int_{-a}^\infty e^{-s(u+a)}H(u)f(u)\,du$$

and given $a > 0 \Rightarrow -a < 0$, the Heaviside function gives the integral as zero on $[-a, 0]$, leaving

$$= e^{-as}\int_0^\infty e^{-su}f(u)\,du = e^{-as}\mathscr{L}[f(t)]\,,$$

given that u acts as a dummy variable. [Bookwork.]

(b) Consider the following second-order ODE

$$\frac{d^2x}{dt^2} + 4\frac{dx}{dt} + 5x = \delta(t-2)$$

satisfying the initial conditions $x(0) = \dfrac{dx}{dt}(0) = 0$, where $\delta(t)$ is the unit-impulse, or Dirac delta-function.

Use the Laplace convolution theorem to show that $x(t)$ satisfies

$$x(t) = 0 \quad \text{for} \quad 0 \le t \le 2\,,$$

and find the form of $x(t)$ for $t > 2$.

SOLUTION Taking Laplace Transforms of the ODE, with zero intial conditions, we have

$$(s^2+4s+5)\bar{x}(s) = e^{-2s} \Rightarrow \bar{x}(s) = \frac{1}{s^2+4s+5}e^{-2s} = \overline{f}(s)\overline{g}(s)$$

We use the first shift-theorem:

$$f(t) = \mathscr{L}^{-1}\left[\overline{f}(s)\right] = \mathscr{L}^{-1}\left[\frac{1}{(s+2)^2+1}\right] = e^{-2t}\mathscr{L}^{-1}\left[\frac{1}{s^2+1}\right] = e^{-2t}\sin t\,,$$

and we already have $g(t) = \delta(t-2)$, so we can employ the convolution theorem, that

$$x(t) = \mathscr{L}^{-1}\left[\overline{f}(s)\overline{g}(s)\right] = f * g = \int_0^t f(t-u)g(u)du = \int_0^t e^{-2(t-u)}\sin(t-u)\delta(u-2)\,du$$

From the properties of the delta function, the last integral is zero if $u = 2$ is not in the interval of integration, hence we require $t > 2$, and for $0 \leq t \leq 2$, the integral is zero, as required. For $t > 2$, the sampling property of the delta function gives

$$x(t) = \int_0^t e^{-2(t-u)}\sin(t-u)\delta(u-2)\,du = e^{-2(t-2)}\sin(t-2)\,, \quad \text{for} \quad t > 2\,,$$

giving the required form of $x(t)$. [Seen similar/new]